

North American Bat Monitoring Program in SE Alaska

Data Summary and Activity Trend Analyses: 2019 - 2025

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1 Executive Summary

The North American Bat Monitoring (NABat) program is a multi-agency initiative administered by the US Geological Survey (USGS). In 2024, the North by Northwest (NNW) Bat Hub was formed by combining the former Alberta Hub and the BC and Southeast Alaska (BC-SE_AK) Hub. The NNW Bat Hub leads the NABat program within S.E Alaska, with key support from state representatives. In 2024, 9 grid cells were monitored and in 2025 6 grid cells were monitored in Southeast Alaska. We used activity, abundance and dynamic occupancy models to evaluate the trends in bat population in the region using 7 years of acoustic data. Results indicate that bat populations across Southeast Alaska remain stable, with some key species showing small positive trends, including Little Brown Myotis and Silver-haired bat. Given growing threats to bat populations across the region, continued monitoring is essential to detect any significant changes.

2 Introduction

2.1 Overview of NABat and the NNW Bat Hub

North American Bat Monitoring Program (NABat) is a large-scale coordinated effort to monitor bat species to address gaps in knowledge and lack of long-term studies across North America (Loeb et al. 2015). NABat uses a spatially-balanced random sampling design and standardized protocols for conducting short-term acoustic surveys and colony counts (Loeb et al. 2015). The continental program is administered by the US Geological Survey (USGS). The Alaska Department of Fish and Game, with assistance from the US Forest Service, initiated an NABat monitoring program in Southeast Alaska in 2019.

NABat relies on regional hubs to help coordinate and standardize monitoring across smaller areas. Regional hubs help facilitate local implementation of NABat and maximize the benefits of the program within each region. The North of Northwest (NNW) Bat Hub was established in 2024 with collaboration from the governments of Alaska, Alberta, and British Columbia. The hub was formed by combining two previous regional hubs: the Alberta Hub and the BC and Southeast Alaska Hub. The NNW Bat Hub is directed by a steering committee consisting of representatives from each jurisdiction, major stakeholders, and USGS team members and it is coordinated by Biodiversity Pathways. The NNW Hub is responsible for compiling data, identifying calls to species, conducting statistical analyses, and generating annual regional reports. NABat monitoring in Southeast Alaska is implemented by the Alaska Department of Fish and Game and US Forest Service.

2.2 Threats to bat populations in S.E. Alaska

North American bat populations are currently facing multiple anthropogenic threats, including logging and other forms of habitat loss and fragmentation, disease, pesticides, wind energy development, and climate change. The most widespread and deadly threat is white-nose syndrome (WNS), a fungal disease that attacks hibernating bats. Although WNS is not yet present in Alaska, it has spread as far west as the Rocky Mountains and as far north as the Northwest Territories in Canada Figure ???. Following its discovery in King County in 2016, WNS has spread throughout most of Washington state and into Oregon Figure ??? and the fungus (though not the disease) has been detected along the Washington border in southern British Columbia (White Nose Syndrome Response Team 2026).

The impacts of white-nose syndrome have been most severe east of the Rocky Mountains, where bats hibernate in large aggregations in caves and mines, resulting in high mortality (Cheng et al. 2021). In the Rocky Mountains and further west, including in Alaska, Little Brown Myotis hibernate singly or in small groups in holes in the soil or between rocks in talus deposits (Neubaum 2018; Blejwas et al. 2021). This very different hibernation behavior is likely responsible for the slower rate of spread of WNS observed in Washington state (Blejwas et al. 2023). Smaller and more dispersed winter colonies may reduce mortality relative to eastern North America, but will also make it difficult to detect the arrival and spread of WNS in Alaska and track its impact on bats in the state (Blejwas et al. 2023). Given the absence of large hibernacula where bats may be counted and swabbed for the presence of WNS, alternative methods for monitoring bats in Southeast Alaska are required to assess changes or impacts from WNS or other threats.

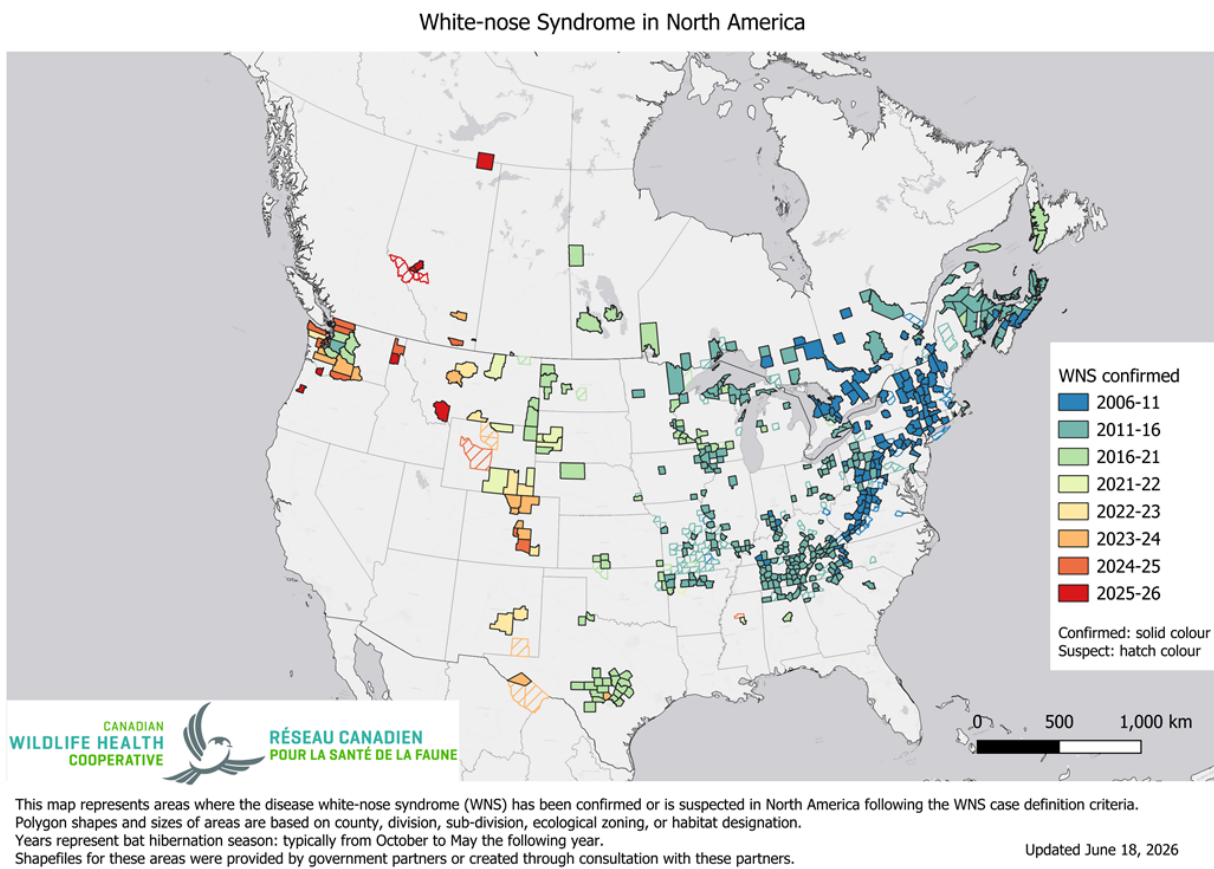


Figure 1: White-nose syndrome (WNS) and *Pseudogymnoascus destructans* (Pd) occurrence in The US and Canada. Source: CWHC-RCSF (2026)

2.3 Acoustic Monitoring of Bats

Acoustic recording is a cost-effective method for monitoring bat populations across large geographic and temporal scales. However, several limitations affect the conclusions that can be drawn from these data and trend estimates cannot be produced for all species using acoustic methods alone:

1. **Imperfect species identification** — Each bat species can produce a broad range of echolocation calls, corresponding to different flight behaviours and environmental features. Species with similar ecology and physiology often produce overlapping calls, particularly in areas with high clutter (obstacles), such as forested habitats. Although detectors are deployed to maximize recordings of diagnostic (typically Search Phase calls), species identifiability will always vary depending on the local species assemblage.
2. **Variable detectability** — Bat species are not equally detectable acoustically. Larger, wide-ranging species produce louder calls that travel farther, whereas species with quieter calls or specialized habitat use may be less reliably recorded.
3. **Data variability** — Many factors influence recording quality and bat activity, including weather, prey abundance and distribution, and local habitat. Additionally, stationary detectors may record an individual bat multiple times, meaning stationary acoustic data can only inform relative activity and occupancy.
4. **Processing limitations** — Auto-ID software enables efficient processing of large datasets, but is known to have high misclassification rates for bats. Manual verification generally produces more accurate results, but has its own limitations, including significantly higher processing time and cost. We use a combination of both approaches, as detailed in the Manual Verification subsection of the Methods.

2.4 Statistical Models

Bat monitoring data can be analyzed using different statistical approaches, each with distinct assumptions and outputs. Activity trend models examine changes in relative acoustic activity over time, providing a straightforward measure of whether bat detections are increasing, decreasing, or remaining stable across the monitoring network.

Dynamic occupancy models take a different approach, estimating the probability that a species occupies a given site, while explicitly accounting for imperfect detection. Rather than treating a non-detection as an absence, these models distinguish between a species being truly absent versus simply going undetected, an important consideration given the variable detectability of bats discussed above. They also track colonization and extinction probabilities across sites over time, providing insight into range shifts and local population dynamics.

In 2025, both model types were run to examine differences in results and assess the complementary information each approach provides.

3 Methods

3.1 Data Collection

3.1.1 Stationary Surveys

Two to four stationary detectors were deployed across at least two quadrants per grid cell for a minimum of seven nights. Sites were selected to maximize species detection and recording quality across different habitat types (see AK report for details). Deployment occurred between mid-May and mid-July, after bats have emerged from hibernation but before young bats are volant. Sites and timing should be kept consistent between years to minimize variability in conditions affecting bat activity.

3.1.2 Transect Surveys

Where possible, driving transects were conducted following Loeb et al. (2015). Bats were recorded along the road by driving at ~32km/h with a microphone mounted vertically inside a magnetic pvc pipe holder on the vehicle's roof. Transects were 30-45 km long and were surveyed twice during the stationary deployment period whenever possible.

3.1.3 Landscape and Weather Covariates

We considered the same covariates for the activity trend models as those previously established by Rae and Blejwas (2026). Temperature and relative humidity were recorded in each grid cell using onsite logging devices where possible; where loggers were unavailable, values were drawn from the nearest climate station. For sites in British Columbia, station records were obtained from the Pacific Climate Impacts Consortium (PCIC) Provincial Climate Data Set (PCDS), which aggregates and quality-controls observations from multiple provincial and federal networks. For sites in Southeast Alaska, station records were obtained using the Meteostat package in Python (Meteostat 2026), which aggregates weather station records from multiple agencies. Additional weather covariates drawn from these station records included nightly mean wind speed (km/h) and nightly precipitation. Degree days above 10°C since January 1 were calculated from daily temperature records, and nightly moon illumination (proportion illuminated $\times$ proportion of the night the moon was above the horizon) was calculated for each site-night.

Site-level covariates were derived from GIS (elevation, distance to water, water body type) or recorded by grid leaders during deployment (distance to clutter, percent clutter); estimates of percent clutter were confirmed using site photos. For sites with smaller ponds that did not appear on the GIS, we used the distance estimate recorded by the grid leader.

We explored additional factors influencing colonization and extinction in the dynamic occupancy models. For those models we included distance to the nearest road, and distance to timber harvest, calculated as the distance from each survey location to the closest road or harvest feature. Road and harvest features for British Columbia were derived from the unpublished Human Footprint Index (HFI) layer currently being developed by the Geospatial Centre within Biodiversity Pathways. Road features for Southeast Alaska were taken from the Alaska Department

of Transportation and Public Facilities ([AKDOT](#)) Roads layer and the US Forest Service Transportation layer. Harvest features for Southeast Alaska were taken from the US Forest Service Tongass_National_Forest_Past_Harvest_from_CoverType layer.

3.1.4 Manual Verification

In 2024 and 2025 acoustic analysis mostly followed the standards set by WCSC in previous years (Rae and Blejwas 2026). This included running all files through two automatic classifiers, Kaleidoscope’s Bats of North America 5.4.0 classifier and Sonobat 3.0’s southeast Alaska classifier, to remove files that do not contain bat pulses and giving a preliminary bat species identification.

Manual vetting largely follows the protocol of Reichert et al. (2018) and previous analysis standards developed by WCSC (Rae and Blejwas 2026). The key difference in the 2024 and 2025 acoustic analysis compared to previous years is that not all automatic species identifications were manually verified. Instead, based on findings from Stratton et al. (2022), at least 50% of recordings were selected for manual verification.

Subsetting for manual verification

Given the project’s large scale, manually vetting all recordings is not feasible long-term. To address this, the NNW Bat Hub, in collaboration with WCSC and Brian Paterson, developed procedures to ensure that selecting a subset of recordings for manual review minimizes biases in analytical results. These standards prioritize the accurate identification of rare and difficult-to-recognize species to maintain the integrity of models and analyses.

To ensure data accuracy, manual verification was conducted based on the number of recordings that received an autoID tag at each site. For sites with 200 or fewer tagged recordings, all non-noise recordings were manually reviewed. For sites with more than 200 tagged recordings, verification depended on the proportion of recordings assigned a species label. All tags for rare species (comprising 10% of prior years’ identifications) and commonly misidentified species were fully vetted. Specifically, all Hoary Bat (*Lasiurus cinereus*) and Silver-haired Bat (*Lasiyonicteris noctivagans*) were reviewed due to the frequent misclassification of noise files that are tagged as these species. Additionally, recordings of common species were randomly selected for verification until 50% of total recordings were reviewed. If fewer than 50% of recordings had species tags, all classifier-assigned tags were manually verified, and “NoID” recordings were randomly selected until 50% of all recordings were reviewed.

Manual verification is complicated by the lack of information on species ranges within Southeast Alaska. We considered all species found in the region as potential candidates in all grid cells with the exception of Long-legged and Yuma Myotis. Long-legged Myotis calls are acoustically similar to those produced by the much more common Little Brown Myotis and auto ID software frequently misclassifies the calls of these two species. Unfortunately, although Little Brown Myotis produces diagnostic calls, Long-legged Myotis does not. Therefore, in grid cells without independent confirmation of the presence of Long-legged Myotis, we downgraded all auto IDs of that species to a Little Brown Myotis/Long-legged Myotis couplet. Where that species has been confirmed present, we retained the species identification unless diagnostic characteristics of Little Brown Myotis were present. Calls of California Myotis also lack diagnostic characteristics and are acoustically similar to Yuma Myotis. However, to date Yuma Myotis has been documented only in the southern portion of Southeast Alaska (Olson and Gunderson 2014) and prior to 2024 we chose to exclude it as a

potential species except for the Ketchikan and Prince of Wales grids. Beginning in 2024, we started including it in the species set for all grids in order to be able to detect any expansion of its range within Southeast Alaska.

3.2 Statistical Analyses

3.2.1 Activity Trends

We conducted species-specific trend analyses using negative binomial GLMM models as described in Rae and Blejwas (2026). We restricted the analysis to data from 2019 onward, due to the low number of grid cells that had been sampled in the years prior (2015 and 2017). We further limited the analysis to grid cells with at least five years of sampling, since estimating a temporal trend requires enough repeat visits per cell to distinguish a directional signal from year-to-year variation.

For each species, we performed model selection by fitting the base model augmented with every combination of up to four covariates, excluding combinations that would introduce collinearity. This produced 387 candidate models per species, from which we selected the model with the lowest AIC as the final covariate set Table ??.

Table 1: Top-performing stationary models selected via AIC-based stepwise selection. Covariates are described as, Distance to Clutter = distance (m) to nearest object; Percent Clutter = estimated percentage of space within a 50m radius obstructed by objects; Water Nearby = detector within 100m of water; WaterDist = distance (m) to nearest water feature; Nightly Temperature = nightly air temperature ($^{\circ}\text{C}$); RH = nightly relative humidity (%); Nightly Mean Wind Speed = mean nightly wind speed (m/s); Nightly Precipitation = total nightly precipitation (mm); jNight = Julian date (day of year) of the survey night.

Species	Species Code	Top Performing Model Covariates
Hoary Bat	LACI	
Silver-haired Bat	LANO	Percent Clutter,Water Nearby,Nightly Min RH
California Myotis	MYCA	Percent Clutter,Nightly Precipitation
Long-eared Myotis	MYEV	Distance to Clutter m ,Nightly Max Temperature,Nightly Min RH,jNight
Little Brown Myotis	MYLU	Percent Clutter,WaterDist,Nightly Precipitation,Nightly Mean Wind Speed
Long-legged Myotis	MYVO	
Yuma Myotis	MYYU	

Post-hoc tests were run on the final models to ensure proper model fit. Species were excluded from the analysis if model fit was not appropriate due to issues including under-dispersion or standard errors that were more than 3 times the estimate. Model results were also excluded if there was less than 150 recordings across all sample years.

3.2.1.1 Use of Automated Identification Results in Stationary Trend Analyses

For stationary trend analyses, species identifications came from the Kaelidoscope Pro AutoID software, not from manual verification. Manual review was used only to remove non-bat noise. Specifically, files that manual review flagged as noise were excluded, but where manual review suggested a different species than the AutoID label, the AutoID classification was retained.

This approach was first identified as necessary in 2023 (Rae and Lausen 2024), when analyses of manually verified data revealed a consistent positive trend across all species, this was interpreted as an artifact of reviewer learning over time rather than a true biological signal. As verifier experience increased across years, calls were increasingly assigned to specific species rather than left unidentified, producing an apparent increase in detections independent of actual bat activity. Because this observer effect is difficult to control for statistically, AutoID classifications are now used as the standard for all activity trend analyses, as automated classifiers apply consistent criteria across all years. While AutoID is subject to some misclassification, the error rate is consistent across years and does not confound detection of genuine temporal trends.

Trend analyses are presented only for species confirmed present through manual verification and for which models fully converged.

3.2.2 Transect Abundance Models

For the transect-based abundance trend models, we used only manually verified detections. When a detection was flagged as a species couplet, it was counted toward each species in the couplet. As above, we restricted the data to transect surveys from 2019 onward and to grid cells with at least five years of sampling. Covariates were limited to weather variables and transect length, and candidate models excluded any combination of collinear variables. We fit 59 candidate models per species and selected the one with the lowest AIC as the final covariate set Table ??.

Table 2: Top-performing transect models selected via AIC-based stepwise selection.

Species	Species Code	Top Performing Model Covariates
Hoary Bat	LACI	Nightly Min Temperature,Nightly Max RH
Silver-haired Bat	LANO	
California Myotis	MYCA	
Long-eared Myotis	MYEV	Nightly Max RH,Nightly Mean Wind Speed
Little Brown Myotis	MYLU	
Long-legged Myotis	MYVO	
Yuma Myotis	MYYU	

After selection, we ran a series of post-hoc checks to confirm model fit. We treated a species' result as reliable only when these tests agreed that the year effect reflected a genuine regional trend rather than an artefact of sparse counts or a single influential cell.

3.2.3 Dynamic Occupancy

We fitted multi-season (dynamic) occupancy models (MacKenzie et al. 2003) to estimate trends in occupancy, colonization, and persistence while accounting for imperfect detection. This analysis used nightly acoustic detection/non-detection data from passive bat detectors deployed at 43 stationary detector sites located in 9 grid cells in Southeast Alaska, as well as 173 stationary detector sites in 56 grid cells in British Columbia (BC). Pooling the two regions allowed the more extensive BC data to inform species-specific detection probabilities and covariate slopes, improving our ability to distinguish true absences from non-detections and thereby sharpening occupancy estimates where Alaska data were sparse. A region effect accounted for structural differences in the landscape between British Columbia and southeast Alaska, allowing us to characterize occupancy trends for southeast Alaska with greater precision than an Alaska-only model would permit.

Data were filtered to the 2017–2025 period, and analysis used an every-other-night subsampling protocol to reduce temporal autocorrelation between consecutive nights, using a maximum of 4 nights per site-year. Sites had to have a minimum of 2 nights of data within a given year and data from at least 2 years total. Models were run separately for all 7 bat species.

Initial occupancy (ψ) was modeled as a function of elevation and water nearby, with a random effect for region. Baseline occupancy for the Alaska region was estimated from Alaska detections only and was not forced toward the BC average. Colonization (γ) and persistence (φ) were modeled with time-varying intercepts and distance to road as a covariate. Detection probability (p) was modeled as a function of percent clutter, nightly mean temperature, and Julian night.

All continuous covariates were standardized by centering on their mean and dividing by their standard deviation; missing covariate values were imputed with the global mean prior to standardization.

Models were fit in a Bayesian framework using JAGS (Version 4.3.2) via the `jagsUI` package (Version 1.6.3) in `r` (Version 4.5.1). We ran 3 chains in parallel for 30,000 iterations each, discarding the first 15,000 as burn-in and thinning the remainder by 5, yielding 9,000 posterior samples for inference. Priors on all regression coefficients were normal with mean 0 and precision 0.368 (equivalent to a standard deviation of approximately 1.65), and the standard deviation of the region-level random effect was assigned a `Uniform(0, 5)` prior. Convergence was assessed using `Rhat` diagnostics, with values below 1.1 taken to indicate convergence. Results are summarized using posterior means and 95% Bayesian credible intervals. We calculated an overall occupancy trend metric () as the ratio of estimated occupancy in the final year to that in the initial year; values of > 1 with 95% credible intervals entirely above 1 indicate an increase in occupancy, values < 1 with intervals entirely below 1 indicate a decline, and intervals overlapping 1 indicate an uncertain or stable.

4 Results

In 2024 all 9 grid cells were monitored, and in 2025 a total of 6 grid sites were sampled in SE Alaska and 3 newly established sites were sampled in the Anchorage area (Figure ??).

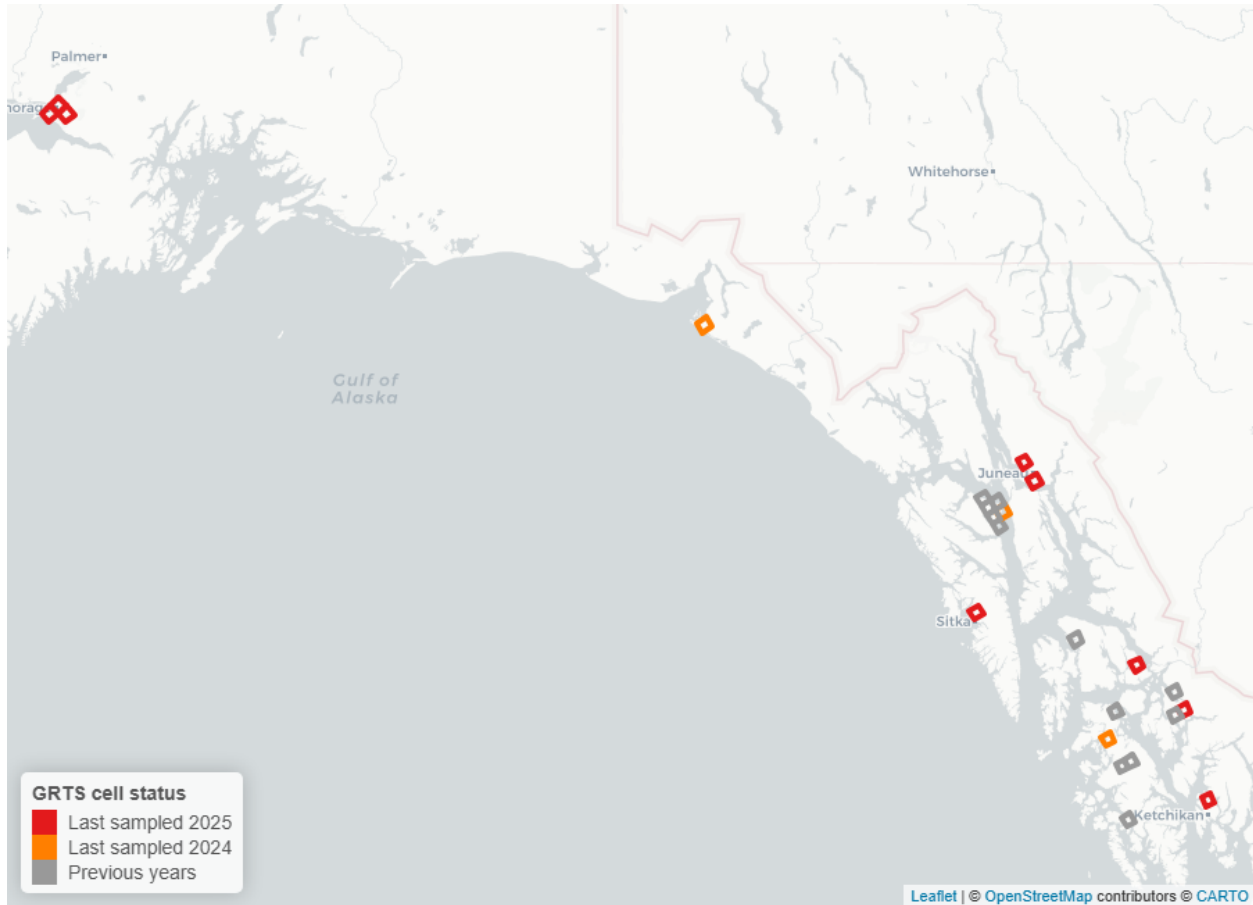


Figure 2: Acoustic bat monitoring sites in Alaska in 2025. NABat grid cells monitored (with two or more stationary sites) in 2025 appear in red and those monitored in 2024 in orange. Cells monitored in previous years but not in 2024 or 2025 appear in grey.

4.1 Activity Summary and Trends

Total acoustic activity varied substantially across sites and years, with Little Brown Myotis dominating call volume at every site in nearly every year. Overall detection totals were highest at the northern sites: Yakutat produced the single largest site-year total (~17,500 calls in 2024), and Juneau recorded consistently high volumes (~12,000 in 2020 and ~13,500 in 2024). Southern and central sites generally recorded lower totals, with Petersburg (~9,000 in 2024) and POW (~7,500 in 2024) reaching comparable peaks in single years, while Wrangell and Hoonah remained low throughout (generally under ~2,000 calls per year). Most sites recorded substantially higher activity in 2024 than in any other year.

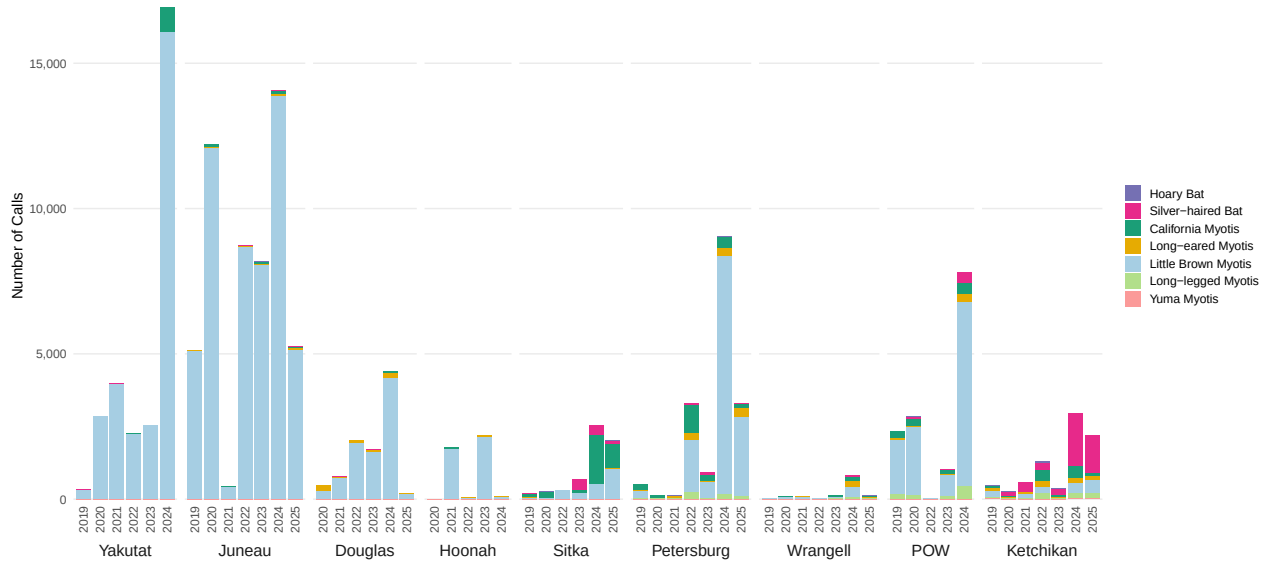


Figure 3: Relative activity of bat species detected during NABat monitoring in 9 grid cells in Southeast Alaska, arranged north to south, during 2019-2025

We estimated yearly trends in nightly acoustic activity for seven species across the stationary detector and mobile transect survey streams. Five species produced reportable stationary-detector models; two (Long-legged Myotis, Yuma Myotis) could not be reliably fit and are not reported Table ???. For transect abundance trends we only report on the results of Silver-haired bat and Little Brown bat since these were the only two species where modelling results were reliable.

Silver-haired Bat showed the clearest and most consistent signal, with activity increasing significantly over the monitoring period in both the stationary ($\beta = 0.48 \pm 0.17$, $p < 0.001$) and transect ($\beta = 0.40 \pm 0.20$, $p = 0.001$) datasets, the only species for which a significant trend was recovered independently in both survey types.

California Myotis showed an increase in activity from the stationary model ($\beta = 0.14 \pm 0.13$, $p = 0.036$). Long-eared Myotis ($\beta = 0.06 \pm 0.13$, $p = 0.35$) and Little Brown Myotis (stationary $\beta = 0.08 \pm 0.14$, $p = 0.24$; transect $\beta = 0.00 \pm 0.12$, $p = 0.94$) showed no detectable trend, with confidence intervals spanning zero in both reportable models.

Hoary Bat, Long-legged Myotis and Yuma Myotis are not reported: Hoary bat and Long-legged Myotis fell below the detection threshold required for trend estimation, and the Yuma Myotis model failed to converge.